

by using a computer. Here's how it works: The pen is filled with a thermo-sensitive ink, which disappears when heated at certain temperature. If you want to erase or edit your work, the computer uses a super-precise laser to heat the specific areas of your drawing that you want to get rid of.

The paper is coated in a light-sensitive photochromic solution, which makes it change colour when exposed to ultraviolet light. Above the workstation, a UV projector can draw on the paper using targeted rays. If you want to copy something you have already drawn, an overhead camera tells the projector to replicate it.

According to the engineers, this technology will truly turn ordinary paper into a display. Until now, it has been possible to project things onto paper and use it as a screen, or import things drawn on paper to a PC by using a digital pen. But the first method uses light, so the results can only be seen in the dark. With the second method, even if you can import things, you can't access them on paper from the computer.

Device that converts images into music

Researchers at the Hebrew University of Jerusalem have come up with EyeMusic—a device that converts images into music. Thanks to pleasant musical tones and scales produced, the device aims to help individuals without vision 'see' using music.

According to a report, once an image is scanned, it represents pixels at high vertical locations as high-pitched musical notes and low vertical locations as low-pitched notes. The image is scanned continuously, from left to right, and an auditory cue used to mark the start of scan. The horizontal location of a pixel is indicated by the timing of musical notes relative to the cue (the later it is sounded after the cue, the farther it is to the right), and the brightness is encoded by the loudness of the sound.

"EyeMusic, which employs pleasant musical scales to convey



(Left) An illustration of EyeMusic SSD, showing a user with a camera mounted on the glasses, and scalp headphones, hearing musical notes that create a mental image of the visual scene in front of him. He is reaching for the red apple in a pile of green ones. (Top right) A close-up of the glasses-mounted camera and headphones; (bottom right) the hand-held camera pointed at the object of interest (Photo credit: Maxim Duply, Amir Amedi and Shelly Levy-Tzedek)

visual information, can be used after a short training period (in some cases, less than half an hour) to guide movements, similar to movements guided visually," explained lead investigators Dr Shelly Levy-Tzedek and Prof. Amir Amedi. "The level of accuracy reached in our study indicates that performing daily tasks with this device is feasible, indicating a potential for rehabilitative use," they added.

Soon HDDs will be able to store five times more

Researchers at Texas University have proved that storage capacity of hard disk drives (HDDs) can be increased by five fold, with the presence of self-organising substances known as block copolymers.

A Crazy Engineer report elaborates the working of this technique: The current production method for HDDs involves writing zeroes and ones as magnetic dots on a continuous surface. The density of these dots determines how much data can be stored on the disk. The closer the dots are, the more the data that can be accumulated. But the proximity of these dots has already maxed out and, in most modern HDDs, the dots are so close that it is impossible to bring them closer without having their magnetic fields interfere with each other.

However, if the dots are isolated from each other so that there is no magnetic field between them, the dots can be pushed together without causing any stability issues. Block copolymers seem to provide an ideal solution to achieve this. These copolymers self-assemble themselves into highly regular patterns of dots and lines. This process, also called directed self-assembly, was pioneered by University of Wisconsin and the Massachusetts Institute of Technology.

The team has managed to break the record by forming the smallest dots of the block copolymers that form the right shapes in a minute. They are expecting to bring the time further down to 30 seconds.

Root any Android device easily with Raspberry Pi based box

Rooting an Android device was never as easy as with the new Raspberry Pi 'root any Android' box. All you need to do is to connect your device to the ingenious box created by XDA developer Adam Outler.

Outler wrote his own scripting language for deploying exploits over the Android Debug Bridge (ADB). Called 'CASUAL' (an abbreviation of 'Cross-platform ADB Scripting Unified Android Loader'), it helps the user to plug in the device and feed exploits